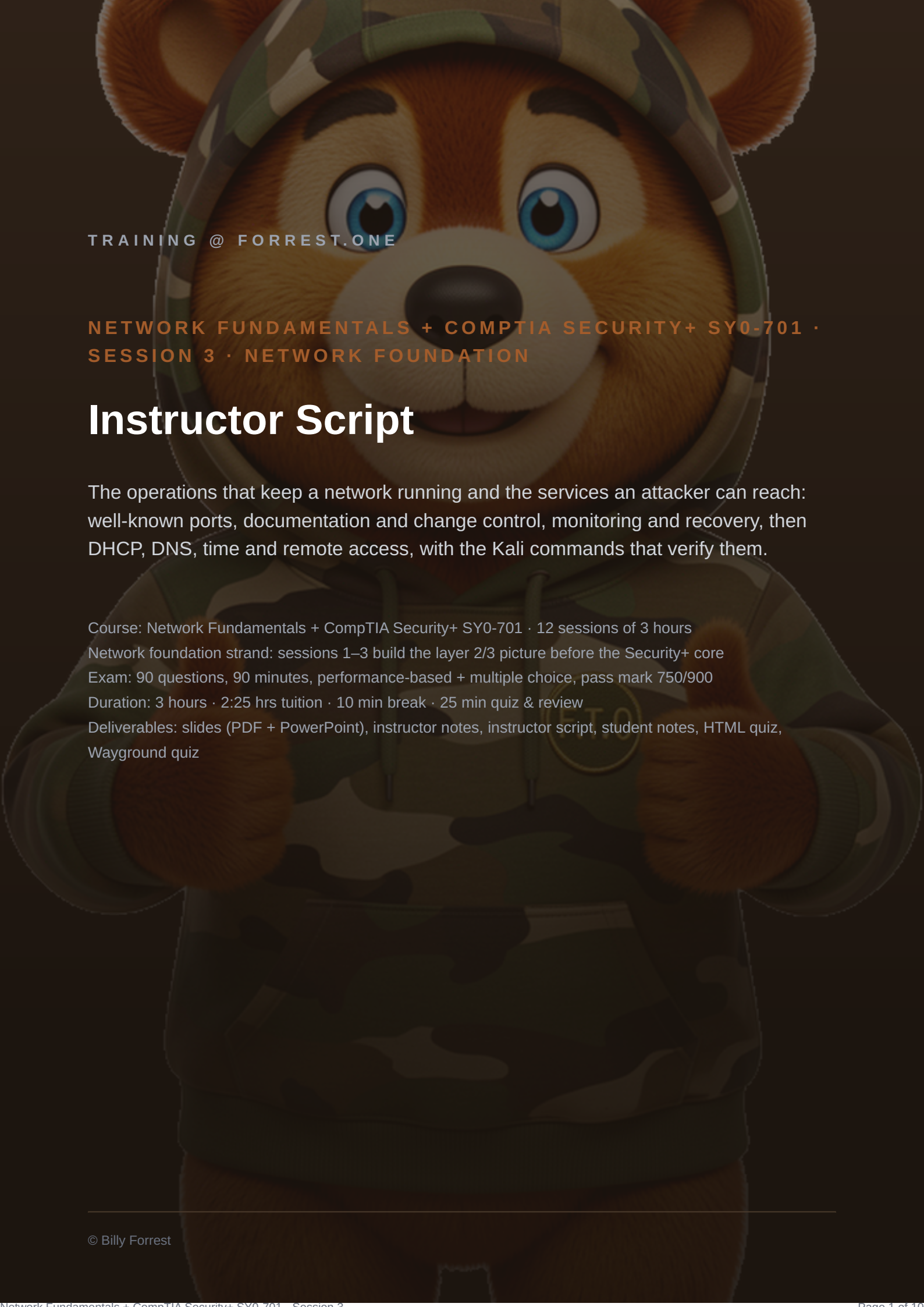




TRAINING @ FORREST.ONE

**NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 3 · NETWORK FOUNDATION**

Instructor Script

The operations that keep a network running and the services an attacker can reach: well-known ports, documentation and change control, monitoring and recovery, then DHCP, DNS, time and remote access, with the Kali commands that verify them.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

Network foundation strand: sessions 1–3 build the layer 2/3 picture before the Security+ core

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Instructor Script: Network Services & the Attack Surface

A word-for-word delivery script for Session 3, the last of the network-foundation strand. Stage directions appear in square brackets. Each slide runs roughly 45 to 90 seconds of speech; activities, discussions and live demos expand the timings in the instructor notes. The script follows the narration: six chapters, in the same order.

Slide 1 · Title

Good day, and welcome to Session Three. Sessions one and two built the map and the roads: models, addressing, switching and architectures. Today we cover the services that run on a network, and the operations that keep them running. We begin with the well-known ports, because every firewall rule and every troubleshooting step starts with a port number. Then we move through documentation, monitoring, disaster recovery and the core services themselves. Keep one question in mind throughout, and that question is what this service lets an attacker do.

Slide 2 · Where Session Three goes

A quick orientation before we start. Chapters one and two cover the well-known ports, the traffic types, the documentation and the agreements. Chapters three to five cover the life cycle, controlled change, monitoring and resilience: the material that decides how fast you recover when something breaks. Chapter six is the core services everything else leans on, addressing, names, time and remote access. This closes the network-foundation strand, because from Session Four onward we study security itself.

Slide 3 · Session objectives

By the end of these three hours you will be able to read a port number, name the service behind it, and say what an attacker gains from it. You will be able to tell a plaintext service from its encrypted replacement, describe the four traffic types, document a network with diagrams, inventories, address management and service levels, control change through its life cycle, monitor with baselines and flow records, choose a recovery site from agreed objectives, explain the core services, and verify all of it on Kali.

Slide 4 · File Transfer and Remote Access

Four services open this section. File Transfer Protocol uses port twenty one to control the session and port twenty to move the data, and neither is encrypted. Secure File Transfer Protocol does the same job inside Secure Shell on port twenty two, so credentials travel protected. Secure Shell gives you an encrypted command line, and it replaced Telnet, which still answers on port twenty three in plaintext. Remote Desktop

Protocol brings a full graphical session on port thirty three eighty nine. Read the port, and you already know the risk. The rule of thumb is simple: if the name has no S, assume plaintext.

Slide 5 · Email, Web and Name Services

Now mail, names and the web. Simple Mail Transfer Protocol relays between servers on port twenty five, but a client submitting its own mail uses port five eighty seven. Domain Name System answers on port fifty three, over User Datagram Protocol for ordinary queries and over Transmission Control Protocol for zone transfers between servers. Dynamic Host Configuration Protocol receives requests on port sixty seven and answers to port sixty eight. The web runs Hypertext Transfer Protocol on port eighty, and the secure version on port four forty three. And remember the troubleshooting rule: names resolve before anything else can happen, so check port fifty three early.

Slide 6 · Management and Communication Ports

Three management protocols remain. Simple Network Management Protocol listens on port one sixty one so a manager can poll a device, and sends its traps to port one sixty two when the device has news. Lightweight Directory Access Protocol answers directory queries on port three eighty nine, and the encrypted form moves to port six thirty six. Session Initiation Protocol sets up voice and video calls on port fifty sixty, with the encrypted signalling on port fifty sixty one. Notice the pattern: the protected form always takes a different port.

Slide 7 · How Traffic Is Addressed

A packet can be addressed four ways. Unicast is one to one, the ordinary case of a browser and a server. Broadcast is one to all on the local segment, which is how a host asks for a hardware address it does not know, and it stays inside one subnet to contain the noise. Multicast is one to many, but only to devices that joined the group, which is why live video scales. Anycast is one to nearest, where many servers share one address and the closest answers. That is how global name servers and content networks scale.

Slide 8 · Discussion

[Run the discussion per the instructor notes.] Which service would you close first on an internet-facing router? Telnet on twenty three and FTP on twenty one go first, because they send credentials in clear text and both have encrypted replacements already in place. Then the harder judgement: SNMP on one sixty one and LDAP on three eighty nine may need to be reachable, but only from management hosts, and SNMP version three should replace the plaintext community string. Every open port is a decision.

Slide 9 · Physical and Logical Diagrams

Chapter two turns to the paperwork that keeps a network honest. Two diagrams answer two different questions, and you need both. A physical diagram shows the devices, the ports in use and the cables between them, so it answers where a cable actually goes. A logical diagram shows the addressing, the virtual local area networks and the traffic flow, so it answers how traffic actually moves. When one workstation loses

its connection, the physical diagram finds the dead port. When everything is slow, the logical diagram finds a route that is too long.

Slide 10 · Specialised Diagrams

Four specialised drawings answer narrower questions. A rack diagram fixes each device to a unit position, so a technician finds it without searching the cabinet, and it records which power feed serves it. One rack unit is one point seven five inches of vertical space. A cable map traces the path of every cable and both of its termination points, which is what makes labelling worth doing. A wiring schematic documents the structured cabling, the patch panels and the distribution frames. A floor plan marks the equipment rooms, the cable tray routes and every wireless access point. Use the drawing to plan the change, not only to record it.

Slide 11 · Asset Inventory and Address Management

You cannot manage what you have not listed. An asset inventory records each device serial number, its location, its configuration and its owner, so a failure has a name and a person attached to it. Software entries carry the version, the licence key and where the software lives. Warranties and service agreements carry end dates, so renewals do not slip past unnoticed. Address management then tracks every static and dynamic allocation, detects conflicts between them, and pushes address changes into the name system, so the list stays true.

Slide 12 · Service Level Agreements

A service level agreement is a contract about performance. It sets an uptime figure, a response time and a resolution time, and the uptime figure is the expensive one. Each additional nine cuts the downtime the business will tolerate roughly tenfold, and the cost of that nine lands on your design, which is why the target decides redundancy, failover and spare capacity. Internal agreements sit between the information technology department and other business units. External agreements cover providers, and a missed target brings service credits or worse.

Slide 13 · Wireless Site Surveys

A site survey is how wireless stops being guesswork. A predictive survey happens before anything is installed: take the floor plan, note the walls and the building materials, model the signal, and decide where the access points should go. A validation survey happens afterwards, measuring real signal strength, noise and throughput to expose the dead zones nobody predicted. The results become heat maps. And one constraint shapes every plan: the lower band offers three non-overlapping channels, one, six and eleven, while the higher band offers many more but reaches less far.

Slide 14 · Life-Cycle and Firmware Management

Chapter three is about ageing equipment and about change. Equipment has a life, and two dates define it. End of Life means the vendor has stopped selling the product, though it may still work well. End of Support is

the cliff, because no further patches arrive, so every flaw found afterwards stays open. Updating is a process rather than an event: download from the vendor, verify the source, stage the image, test it away from production, then deploy in a maintenance window. Keep similar devices on one version, and keep a tested rollback plan.

Slide 15 · Asset Decommissioning

Retiring a device is a security event, not a bin run. A router or a switch still holds its configuration, its certificates and its stored credentials, so a factory reset and an erase come first. For storage media, physical destruction gives the strongest assurance, degaussing suits magnetic drives, and overwriting suits less sensitive data. Then the paperwork: certified recycling to meet electronic waste rules, and updates to the inventory, the diagrams and the monitoring, so nothing alerts on a device that is already gone.

Slide 16 · Controlled Change

Most outages are self-inflicted, and change control is the antidote. A Change Advisory Board reviews anything that can disrupt service. Standard changes are routine and pre-approved, normal changes are planned, tested and reviewed, and emergency changes go in first and face the review afterwards. Every request carries a justification, a risk assessment, an impact analysis, a rollback plan and a schedule agreed with the people affected. A post-implementation review then records what worked and what did not, and the audit trail keeps the whole story.

Slide 17 · The document you did not update is the one you will trust

One habit ties this chapter together. Every change touches four records: the network diagram, the asset inventory, the configuration baseline and the monitoring configuration. Skip one of them, and the next engineer works from fiction. A diagram that disagrees with reality is worse than no diagram at all, because it earns confidence it has not deserved. So make documentation part of the change itself, with a named owner, rather than a chore for later.

Slide 18 · Configuration Management and Backups

Chapter four moves from paperwork to telemetry. Configuration management is about predictability. The production configuration is what a device runs now, and the baseline configuration is the approved template for that device type. Drift is the slow gap between the two, opened by a manual tweak nobody recorded. Compliance checking finds it by comparing the running settings against policy. Back up by hand, on a schedule, or automatically when a change is detected, and keep copies locally, in version control and off site. A backup you have never restored is a hope, not a plan.

Slide 19 · Simple Network Management Protocol

Simple Network Management Protocol asks a device how it feels. A manager polls an agent, and the agent answers from its management information base. Every value has an object identifier, so the manager can

request exactly the counter or the status it wants. GET reads a value, SET changes one, and a trap is the unsolicited alarm when something crosses a threshold. Requests go to port one sixty one and traps to port one sixty two. Older versions send a community string in plaintext, so treat that string as a password in the clear, and use version three.

Slide 20 · Flow Data

Flow data gives you the wide shot. A flow record describes a conversation: who talked to whom, on which ports, using which protocol, and how many bytes. NetFlow collects those records as traffic crosses an interface. sFlow samples packets at intervals instead, which keeps it workable on fast links. The Internet Protocol Flow Information Export standard formalises the same idea. Because a flow is only a summary, it costs far less storage than full packets, so you can keep it for a long time.

Slide 21 · Flow Records or Full Capture

So which one do you reach for? Start with flow records, because they are cheap to keep and they show the shape of the day, which is how floods show up. Move to full capture when you need to read what was actually said, such as a malformed exchange or a suspicious payload. Capture needs a mirror port or a passive test access point, filters, and real storage. And remember the legal ground: packets hold personal data, so capture only with authority.

Slide 22 · Baselines and Metrics

Chapter five asks what normal looks like, and what happens when normal ends. A baseline is what normal looks like, written down. You record bandwidth utilisation, latency, error rate and device load under ordinary conditions, and from that point a number means something. Threshold alerting fires when a fixed limit is crossed, which is simple and prone to false alarms. Anomaly detection instead watches the shape of the traffic and flags what departs from the baseline. Trending turns those records into a forecast for capacity, and remember that a baseline is only valid until the network changes around it.

Slide 23 · Logs and Integration

Logs are only useful if you can find them. Log aggregation pulls entries from routers, switches and firewalls into one searchable repository, so an incident has a single story. Syslog carries each message with its source, its timestamp and a severity, from zero, an unusable system, down to seven, which is chatter. A security information and event management platform correlates across sources and surfaces a pattern no single device shows. Interfaces pull cloud logs into the same place, and automation can block a source as a rule fires.

Slide 24 · Disaster Recovery Metrics

Disaster recovery begins with numbers the business must choose. The Recovery Point Objective says how much data you can afford to lose, and it sets how often you back up. The Recovery Time Objective says how

long you can afford to be down, and it decides how much failover you buy. Mean Time To Repair measures how quickly you restore service, and Mean Time Between Failures measures how long a component runs before it fails. Low numbers cost money, so agree them with the business. Do not invent them.

Slide 25 · Disaster Recovery Sites

Where you recover is a trade between cost and speed. A cold site gives you space, power and connectivity, so it is cheap and slow, since everything is bought first. A warm site has some equipment ready, so it comes up faster at a higher cost. A hot site mirrors production continuously, so you switch across almost at once, which is why it costs most. Place it far enough away to miss the same flood, close enough to keep replication quick, and inside the rules on where data may sit.

Slide 26 · High Availability Architectures

High availability is about surviving a failure without the users noticing. In an active-active design every node serves traffic, so the load is shared and one loss barely shows, though each node must carry capacity for the rest. In an active-passive design a standby waits, watching a heartbeat, and takes over only when the heartbeat stops. A load balancer spreads requests across healthy nodes and steers away from any that stop answering. Spread nodes across sites to survive a regional failure, and accept that faster recovery costs more capacity.

Slide 27 · Disaster Recovery Testing

A plan that has never been tested is a guess with a cover page. A tabletop exercise walks the plan through as a discussion, which exposes wrong assumptions cheaply. A simulation has the team perform the recovery tasks away from production. A parallel test runs the recovery systems alongside live traffic, to prove they can carry the real load. A full failover switches everything across, which is the only complete proof and the most disruptive. Document the results, fix what failed, and repeat at least once a year.

Slide 28 · Break

Ten minutes. When we return, the core services everything leans on: dynamic addressing, names, time and remote access, and the commands that verify each one.

Slide 29 · Dynamic Addressing

Chapter six is the services everything else leans on. Dynamic Host Configuration Protocol hands out the settings a device needs to join. The client broadcasts a discovery from no address at all, servers offer, the client requests one, and the server acknowledges it. That exchange is known as DORA, and you should recite it from memory. Where the server sits on another subnet, a relay agent carries the conversation across. A reservation pins one address to one hardware address, scope options deliver the gateway and name servers, and two servers in failover can share the pool. And keep the risk in mind: the client accepts the first offer it receives.

Slide 30 · Automatic Addressing in Version Six

Internet Protocol version six changes who configures addresses. A router multicasts an advertisement carrying the network prefix, and the host combines it with an interface identifier to build its own address, a method called Stateless Address Autoconfiguration. That identifier can come from the hardware address, with F F F E inserted in the middle. Where an administrator wants a record of every address, stateful DHCP version six assigns and tracks them, or stateless DHCP version six supplies settings alone. Privacy extensions rotate temporary addresses so a device cannot be tracked.

Slide 31 · Name Services

Domain Name System is the service everything leans on. Thirteen sets of root servers point each query down to the top level domains, which point on to the authoritative servers holding the real records. Behind a name sit an address record for version four, a quad A for version six, aliases, mail exchangers, reverse pointers and text records for mail policy. Each record carries a time to live, which decides cache lifetime. Split horizon answers internal and external clients differently, and the security extensions sign records, so forgery fails.

Slide 32 · Time Synchronisation Services

Every log entry is only as trustworthy as the clock that wrote it. The Network Time Protocol organises time into strata. Stratum zero is the reference clock, often atomic or fed by satellites, and stratum one servers sit directly on it, with the layers descending from there so the load is spread. Configure several sources, because that gives redundancy and cross checking. When time drifts, tickets expire, logs disagree and an incident timeline falls apart, so authenticate the traffic and stop a forged source shifting your clock.

Slide 33 · Virtual Private Networks

Virtual private networks carry private traffic over public links. A site to site tunnel joins two offices, while a remote access tunnel serves one user through client software. Internet Protocol Security encrypts at the packet level, in transport mode for the payload and tunnel mode for the whole packet. Full tunnelling sends everything through the tunnel, and split tunnelling sends only corporate traffic, which saves bandwidth but blurs the boundary you are defending.

Slide 34 · Remote Access and Device Management

To manage the equipment itself, you have four paths. Use a console cable when nothing else answers, because it is out of band and needs no network. Use Secure Shell for routine work, over an encrypted command line. Use the web interface for simple tasks and less experienced staff. Use a programmatic interface when you need automation and scale. Every one of these paths is an attack surface, so harden all four and turn off the ones you do not use.

Slide 35 · Verifying services on Kali

[Run the commands live.] DNS: dig bbc.co.uk plus short, and cat the resolv configuration file. DHCP: cat the dhclient lease file to see the lease history. Time: ntpq -p for peers and offset, timedatectl for the clock, journalctl for the NTP events. SSH: ss -tnp piped to grep for port twenty two, and grep sshd in the authentication log. And the master command: ss -tuln shows everything listening. An analyst should be able to explain every listener on the box, or admit there is a gap.

Slide 36 · Reading the evidence

[Run the forensic commands live.] Failed password count: grep for failed passwords in the authentication log, piped to a word count, gives the brute-force volume. Accepted logins: grep for accepted entries. Active sessions: ss with the process flag, filtered for ports twenty two, fifty three and one twenty three. The analyst habit: correlate SSH and RDP session times with user accounts and network flows for a complete picture. And the red flag: an unexpected service listening on a high port is an indicator of compromise.

Slide 37 · Quiz

[Direct candidates to Quiz_Session_03.html. 15 minutes.] Open the session quiz. Twenty questions, immediate feedback, and every question is tagged with its topic area, so you will see the tag, like 3A DNS, above each question. Read every explanation; the distractors teach as much as the answers. Aim for seventy percent or better before you leave today. The Wayground version is available for in-class assessment.

Slide 38 · Key takeaways

Six takeaways. Ports come in pairs, one plaintext and one encrypted, so learn them together. Traffic is unicast, broadcast, multicast or anycast. Keep both diagrams current, because a wrong drawing is worse than none. Inventories, address management and service levels decide how quickly you recover. End of Support is the security cliff, so patch with a rollback ready and document the change. Recovery objectives choose your site, and an untested plan is only a guess. And harden every management path.

Slide 39 · Revision checklist

Before next session: learn the port pairs, twenty one with twenty two, twenty three with twenty two, twenty five with five eighty seven, fifty three, sixty seven with sixty eight, eighty with four forty three, one sixty one with one sixty two, three eighty nine with six thirty six, and fifty sixty with fifty sixty one. Monitor with flow records first and capture packets only when you must. Remember that dynamic addressing, names and time are the services everything leans on. Run the commands on any Linux box you can reach, and retake the quiz until you score seventy percent without guessing.

Slide 40 · Next session preview

So where does this leave you? You can read a port number, name the service behind it, and say what an attacker gains from it. You can document a network, control a change, watch a baseline, choose a recovery site and prove that the plan works. Session Four begins the security core: confidentiality, integrity and availability, the categories of control, and the framework that organises them. Take one habit from today. Before you change anything, ask what breaks if it fails, and what you will do when it does.